CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

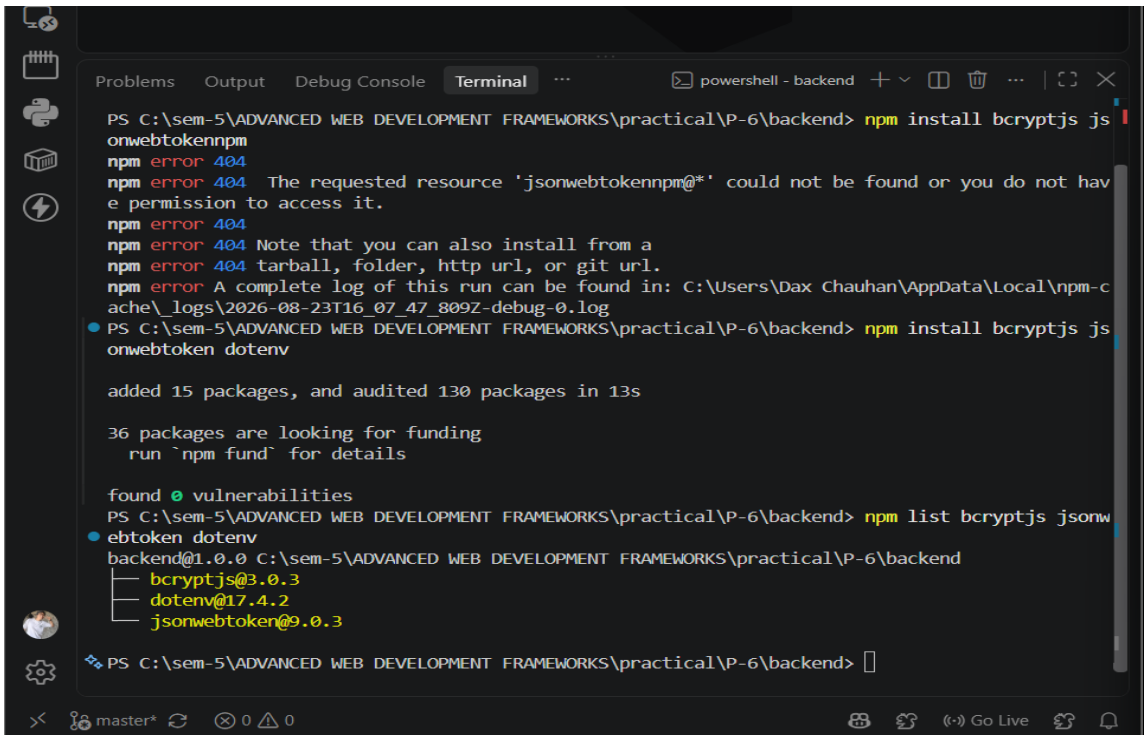
B. TECH (IT/CE/CSE/AIML)

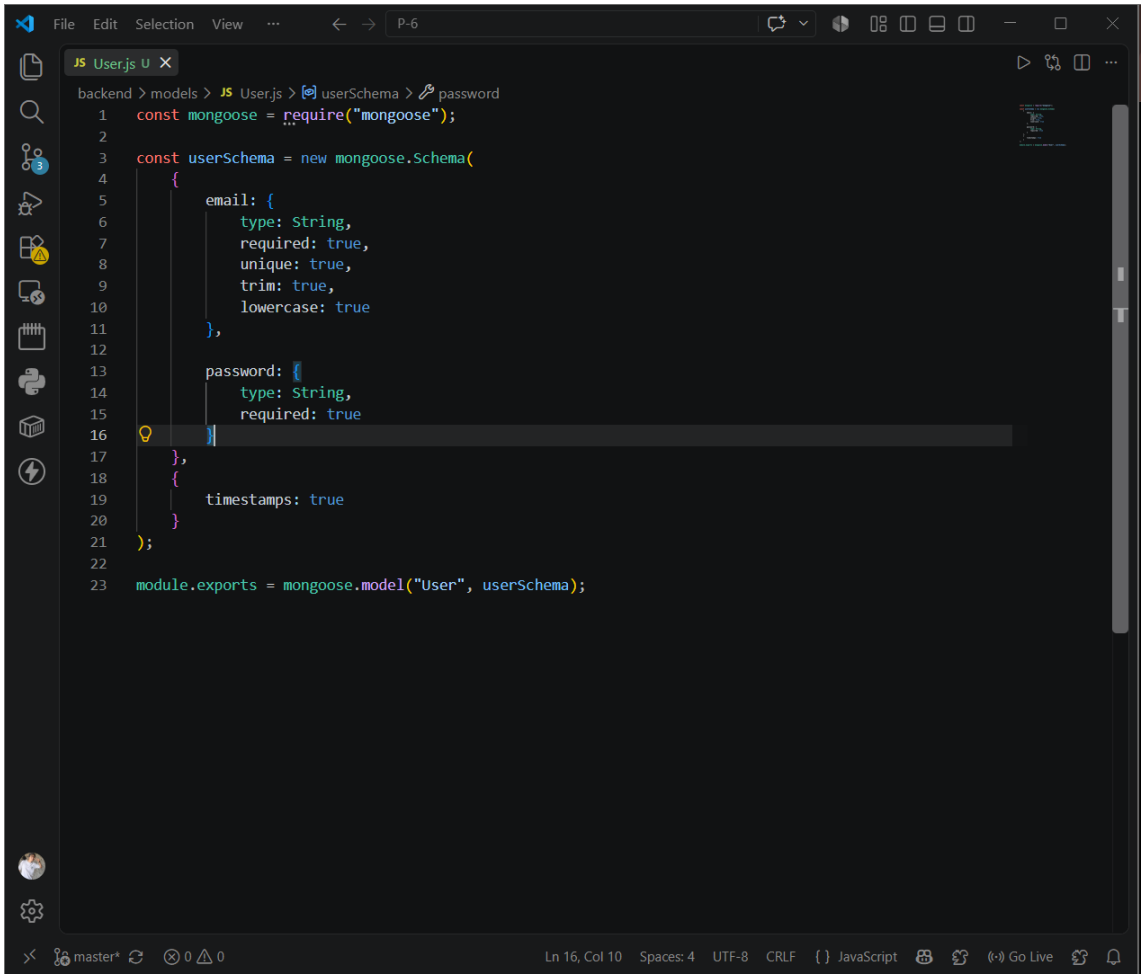
Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

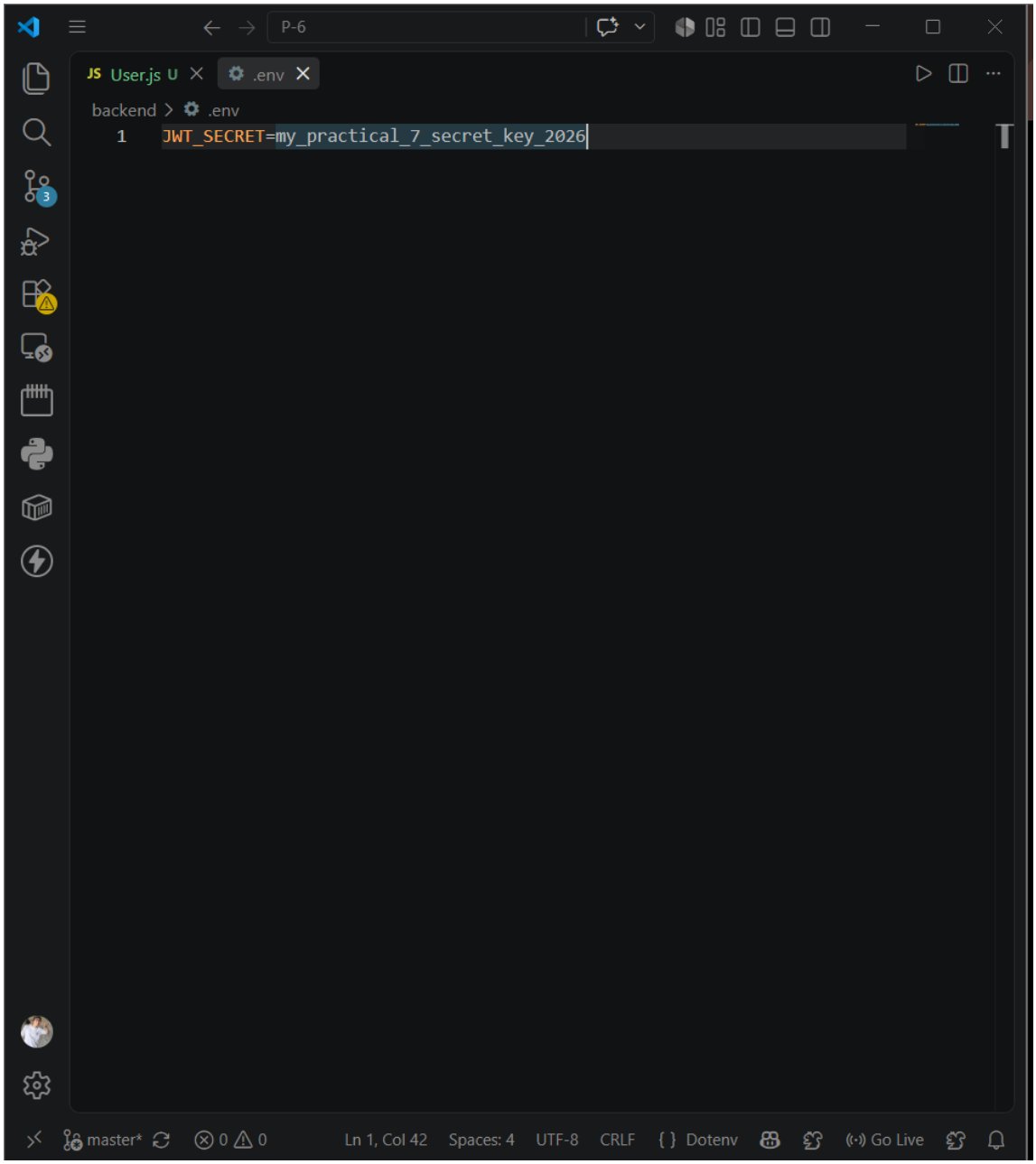
Practical Statement:

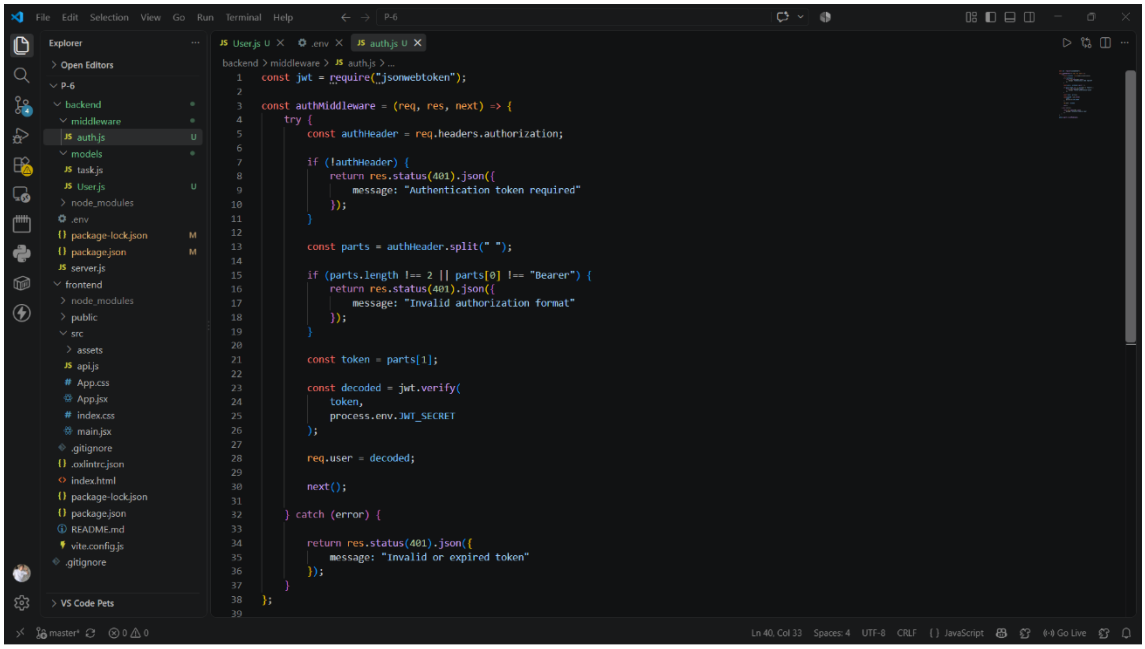
Develop a secure full-stack Task Management application that implements **user authentication and authorization using JWT**. The system should allow users to register and log in securely, hash passwords using **bcrypt**, generate JWT tokens after successful login, and protect task-related API routes using authentication middleware. The application should also perform server-side validation of task data and handle invalid or expired tokens without crashing the server.

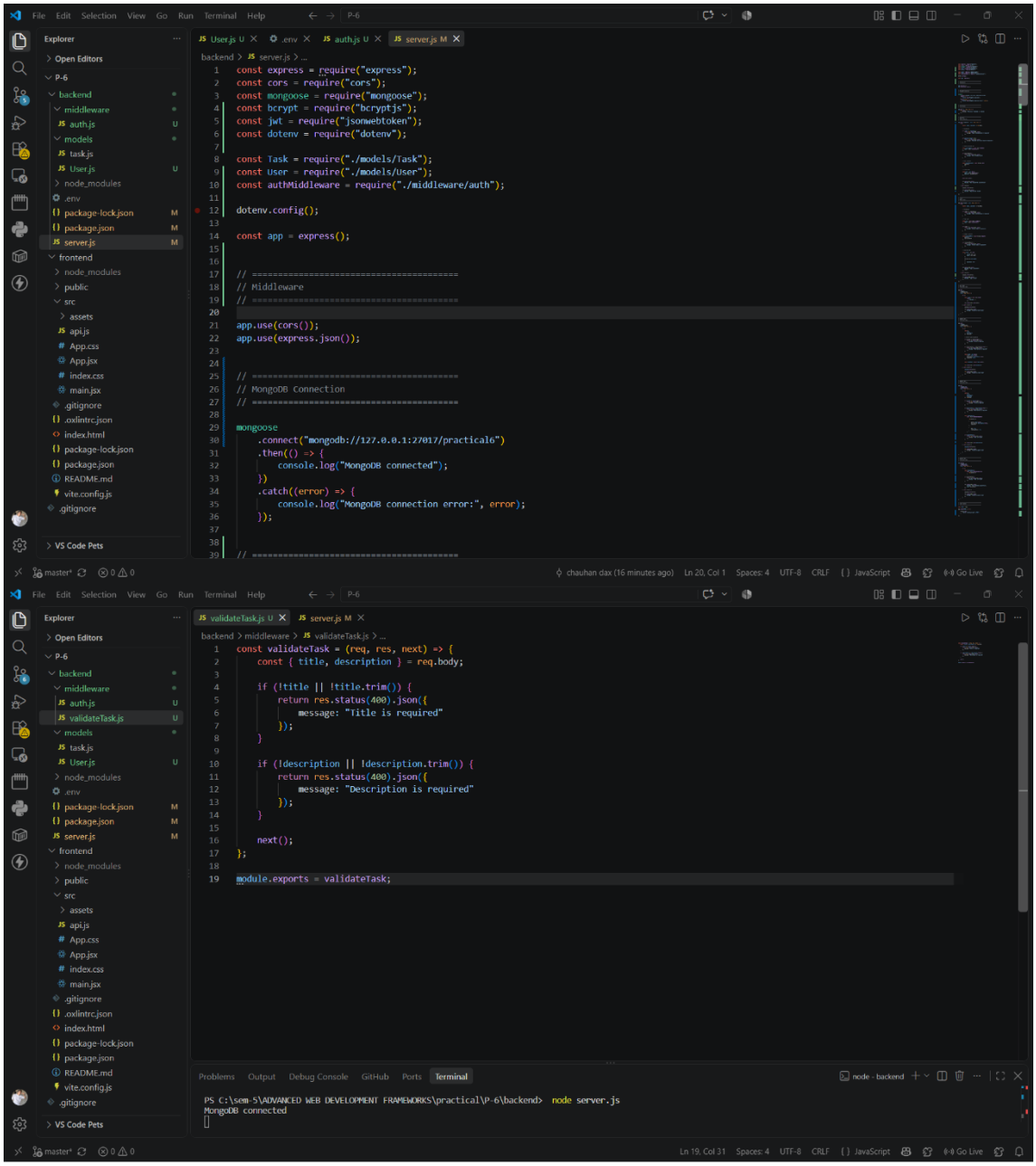
Screenshots:

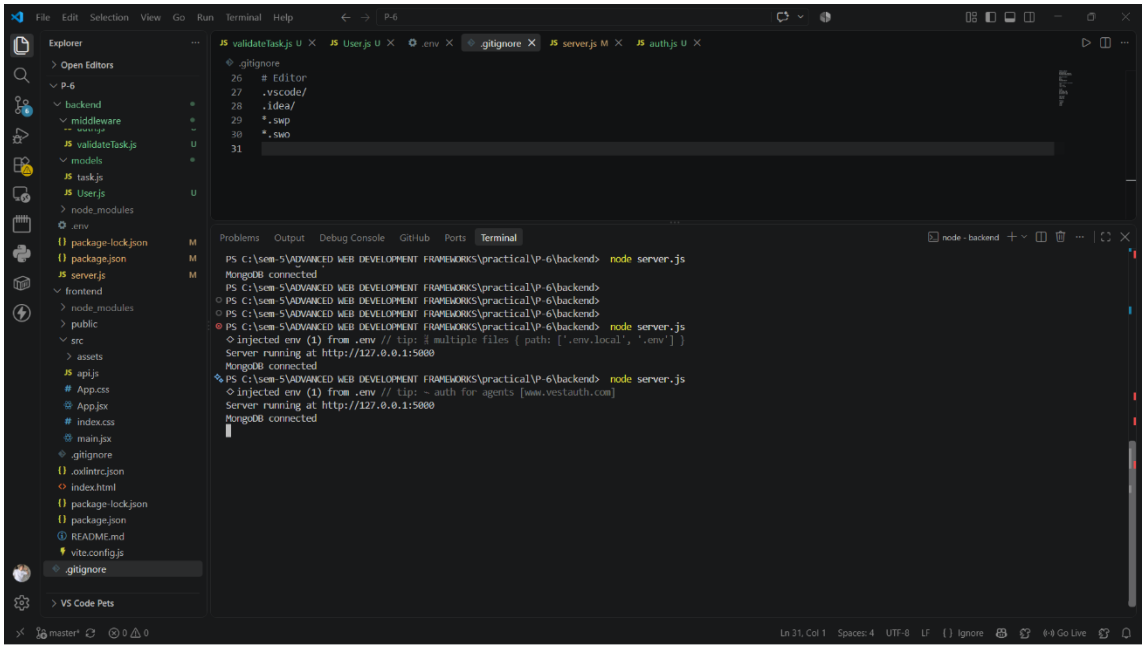


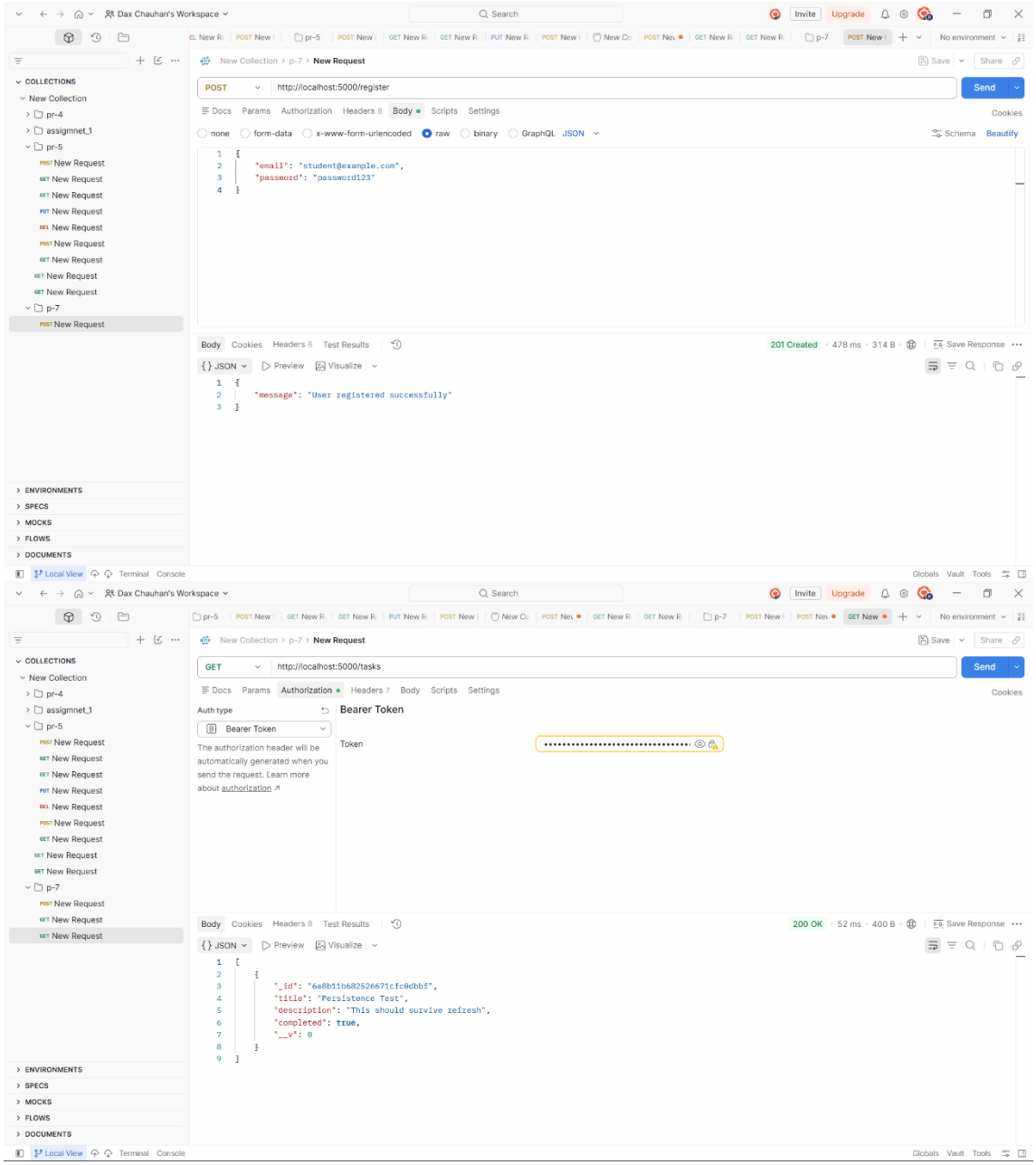

File Edit Selection View ···
P-6
User.js U
backend > models > User.js > userSchema > password
const mongoose = require("mongoose");
const userSchema = new mongoose.Schema(
{
email: {
type: String,
required: true,
unique: true,
trim: true,
lowercase: true
},
password: {
type: String,
required: true
}
},
{
timestamps: true
}
);
module.exports = mongoose.model("User", userSchema);
master*
Ln 16, Col 10
Spaces: 4
UTF-8
CRLF
{ } JavaScript
Go Live

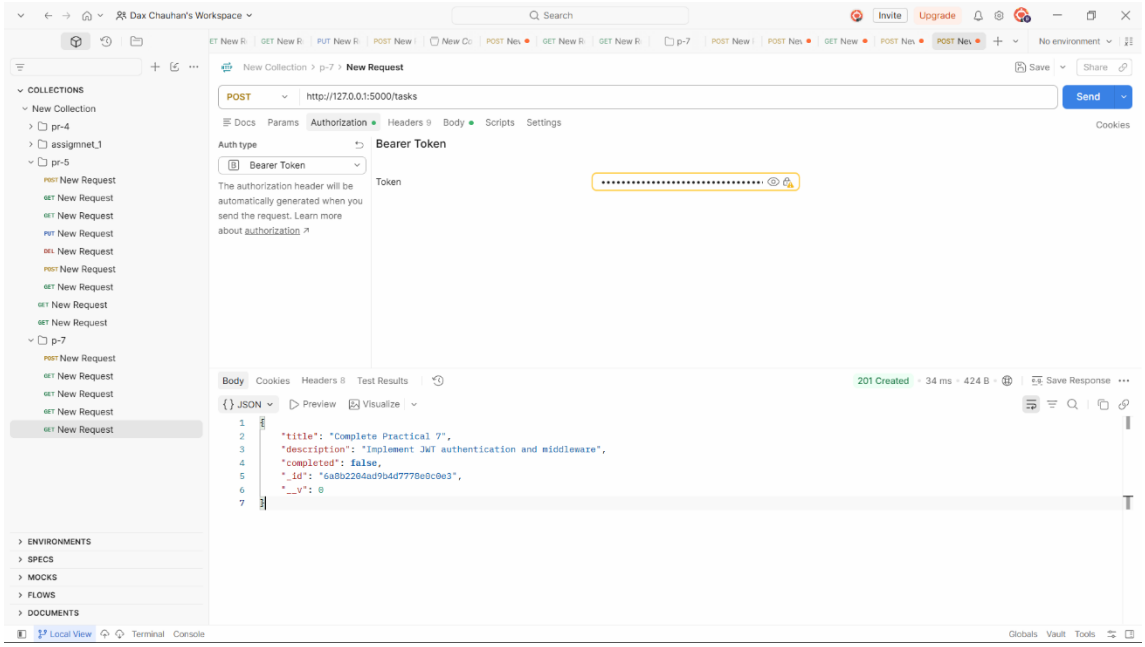

P-6
User.js U
.env
backend > .env
1 JWT_SECRET=my_practical_7_secret_key_2026
master*
0 0
Ln 1, Col 42
Spaces: 4
UTF-8
CRLF
{ } Dotenv
Go Live

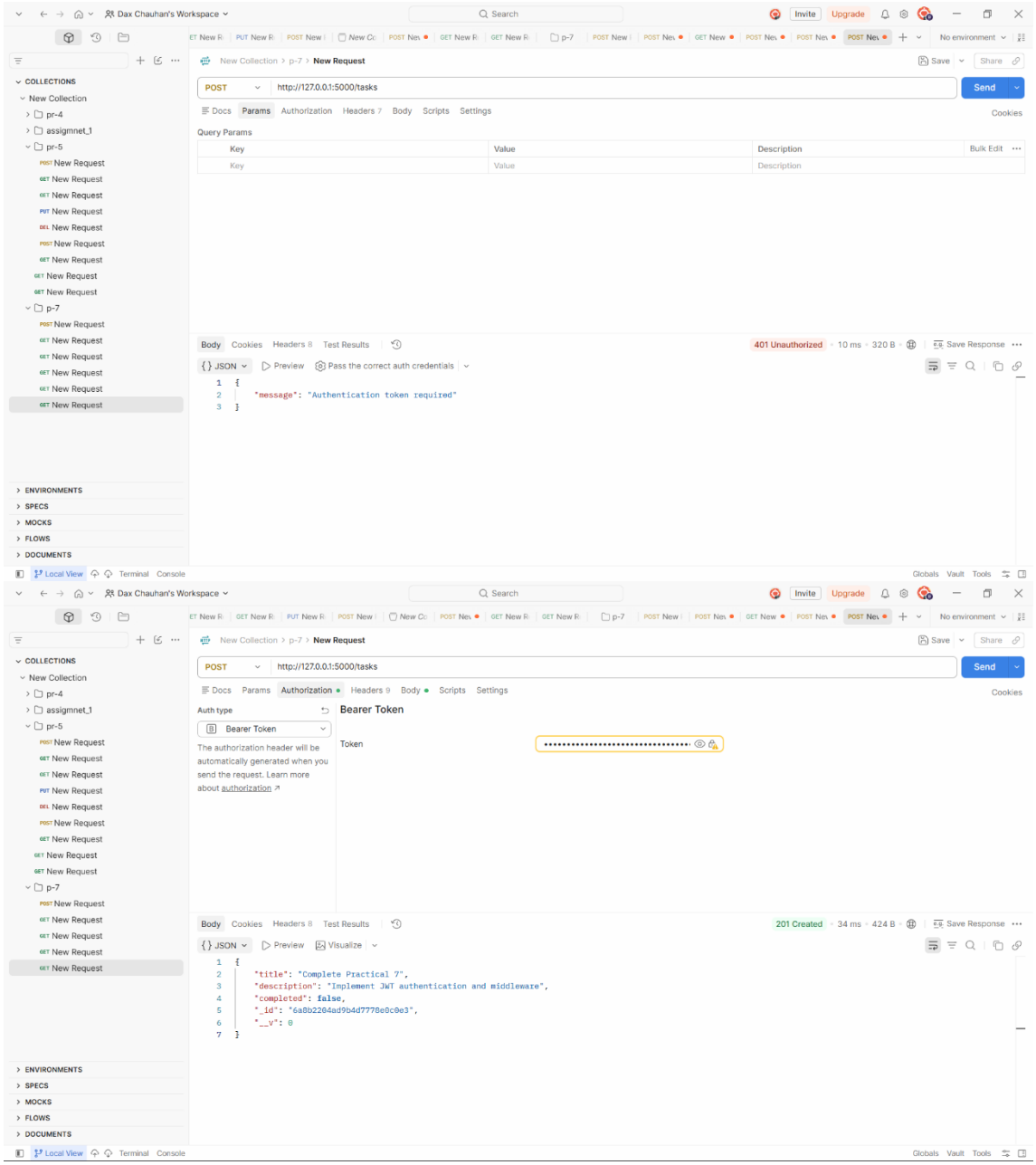

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Explorer
Open Editors
P-6
backend
middleware
auth.js
models
task.js
User.js
node_modules
.env
package-lock.json
package.json
server.js
frontend
node_modules
public
src
assets
api.js
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
README.md
vite.config.js
.gitignore
VS Code Pets
User.js U
.env
auth.js U
backend > middleware > auth.js > ...
const jwt = require("jsonwebtoken");
const authMiddleware = (req, res, next) => {
try {
const authHeader = req.headers.authorization;
if (!authHeader) {
return res.status(401).json({
message: "Authentication token required"
});
}
const parts = authHeader.split(" ");
if (parts.length !== 2 || parts[0] !== "Bearer") {
return res.status(401).json({
message: "Invalid authorization format"
});
}
const token = parts[1];
const decoded = jwt.verify(
token,
process.env.JWT_SECRET
);
req.user = decoded;
next();
} catch (error) {
return res.status(401).json({
message: "Invalid or expired token"
});
}
};
master*
Ln 40, Col 33
Spaces: 4
UTF-8
CRLF
{ } JavaScript
Go Live

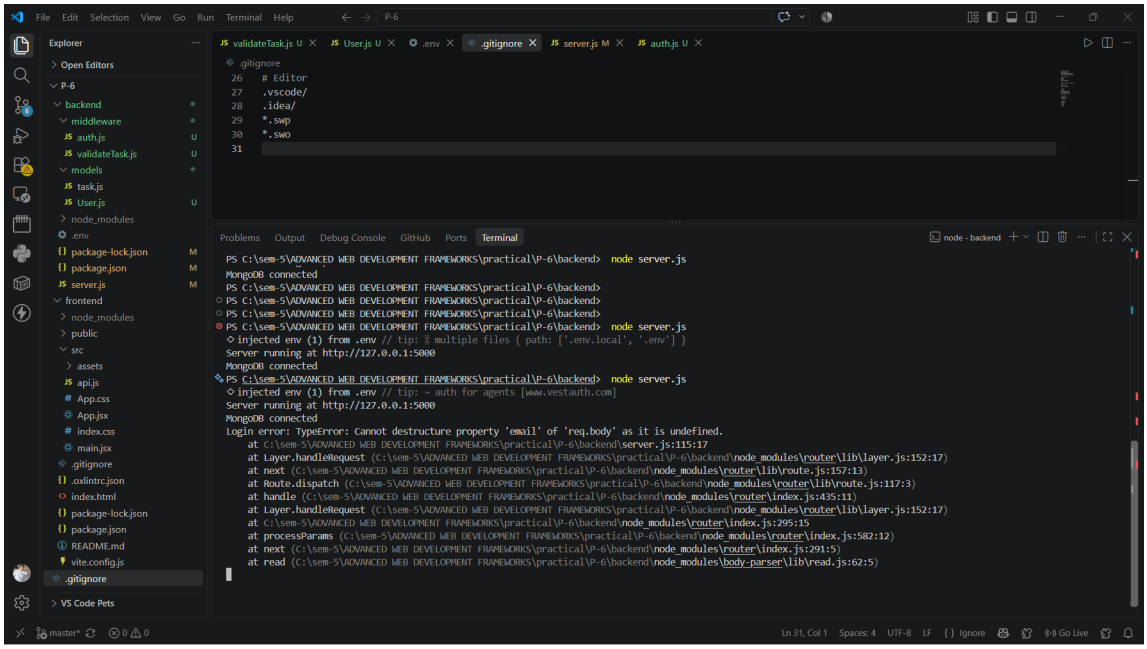

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middleware
auth.js
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package-lock.json
package.json
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node_modules
public
src
assets
api.js
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
README.md
vite.config.js
.gitignore
VS Code Pets
User.js U
.env
auth.js U
server.js M
backend > server.js > ...
const express = require("express");
const cors = require("cors");
const mongoose = require("mongoose");
const bcrypt = require("bcryptjs");
const jwt = require("jsonwebtoken");
const dotenv = require("dotenv");
const Task = require("./models/Task");
const User = require("./models/User");
const authMiddleware = require("./middleware/auth");
dotenv.config();
const app = express();
// ======================================
// Middleware
// ======================================
app.use(cors());
app.use(express.json());
// ======================================
// MongoDB Connection
// ======================================
mongoose
.connect("mongodb://127.0.0.1:27017/practical6")
.then(() => {
console.log("MongoDB connected");
})
.catch((error) => {
console.log("MongoDB connection error:", error);
});
// ======================================
chauhan dax (16 minutes ago) Ln 20, Col 1 Spaces: 4 UTF-8 CRLF { } JavaScript Go Live
validateTask.js U
server.js M
backend > middleware > validateTask.js > ...
const validateTask = (req, res, next) => {
const { title, description } = req.body;
if (!title || !title.trim()) {
return res.status(400).json({
message: "Title is required"
});
}
if (!description || !description.trim()) {
return res.status(400).json({
message: "Description is required"
});
}
next();
};
module.exports = validateTask;
Problems Output Debug Console GitHub Ports Terminal
node - backend
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend> node server.js
MongoDB connected
Ln 19, Col 31 Spaces: 4 UTF-8 CRLF { } JavaScript Go Live


File Edit Selection View Go Run Terminal Help
Explorer
Open Editors
P-6
backend
middleware
validateTask.js
models
task.js
User.js
node_modules
.env
package-lock.json
package.json
server.js
frontend
node_modules
public
src
assets
api.js
App.css
App.jsx
index.css
main.jsx
.gitignore
.oxlintrc.json
index.html
package-lock.json
package.json
README.md
vite.config.js
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VS Code Pets
validateTask.js U
User.js U
.env
.gitignore
server.js M
auth.js U
.gitignore
26 # Editor
27 .vscode/
28 .idea/
29 *.swp
30 *.swo
31
Problems Output Debug Console GitHub Ports Terminal
node - backend
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend> node server.js
MongoDB connected
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend>
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend>
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend>
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend> node server.js
injected env (1) from .env // tip: ⌘ multiple files { path: ['.env.local', '.env'] }
Server running at http://127.0.0.1:5000
MongoDB connected
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\P-6\backend> node server.js
injected env (1) from .env // tip: ~ auth for agents [www.vestauth.com]
Server running at http://127.0.0.1:5000
MongoDB connected
master*
Ln 31, Col 1 Spaces: 4 UTF-8 LF { } Ignore
Go Live


POST
http://localhost:5000/register
Send
{
"email": "student@example.com",
"password": "password123"
}
201 Created
{
"message": "User registered successfully"
}
GET
http://localhost:5000/tasks
Send
Bearer Token
Token
200 OK
[
{
"_id": "6a8b11b682526671cfc0dbbf",
"title": "Persistence Test",
"description": "This should survive refresh",
"completed": true,
"__v": 0
}
]


Dax Chauhan's Workspace
Search
Invite
Upgrade
No environment
New Collection > p-7 > New Request
Save
Share
COLLECTIONS
New Collection
pr-4
assignment_1
pr-5
p-7
POST
http://127.0.0.1:5000/tasks
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
Auth type
Bearer Token
Token
The authorization header will be automatically generated when you send the request. Learn more about authorization
Body
Cookies
Headers 8
Test Results
201 Created
34 ms
424 B
Save Response
JSON
Preview
Visualize
{
"title": "Complete Practical 7",
"description": "Implement JWT authentication and middleware",
"completed": false,
"_id": "6a8b2204ad9b4d7778e8c0e3",
"__v": 0
}
ENVIRONMENTS
SPECS
MOCKS
FLOWS
DOCUMENTS
Local View
Terminal
Console
Globals
Vault
Tools


POST
http://127.0.0.1:5000/tasks
Send
401 Unauthorized
{
"message": "Authentication token required"
}
Bearer Token
Token
201 Created
{
"title": "Complete Practical 7",
"description": "Implement JWT authentication and middleware",
"completed": false,
"_id": "6a8b2264ad9b4d7778e0c0e3",
"__v": 0
}



Github repo: https://github.com/sahilmak1602/WDF_24IT045_SAHIL.git